

2. The diagram shows regions of the electromagnetic (em) spectrum.
The regions are in order.

radio waves	microwaves	infra-red	visible light	ultraviolet	X-rays	gamma rays
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- (a) **Tick (✓) the three correct statements.** [3]

The frequency of em waves decreases from radio waves to gamma rays. ☐

All em waves travel at the same speed in a vacuum. ☐

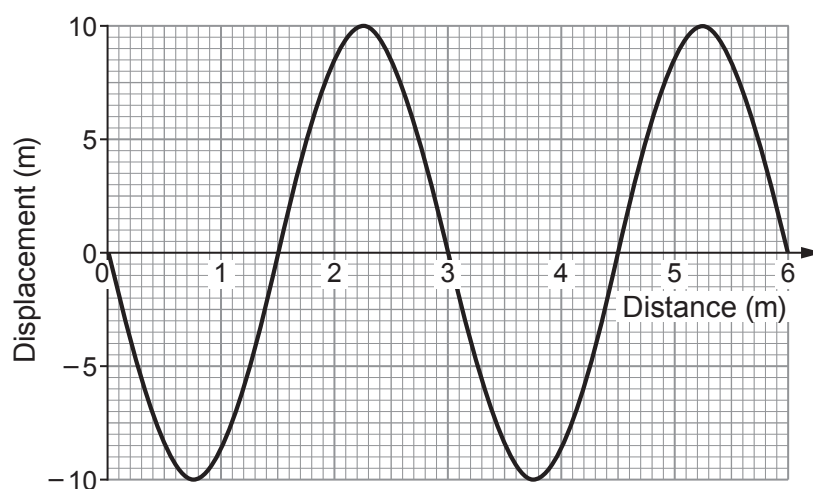
Gamma rays can be emitted from radioactive materials. ☐

The wavelength of em waves decreases from radio waves to gamma rays. ☐

Radio waves, microwaves, infra-red and visible light are all ionising radiations. ☐

All em waves are longitudinal. ☐

- (b) The graph below shows a wave diagram.



Use the following values to answer the questions below. You may use each value once, more than once or not at all.

2	3	5	6	10
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- (i) I. State the amplitude of the wave. m [1]
 II. State the number of complete waves shown. waves [1]
 III. State the wavelength of the wave. m [1]



- (ii) I. Complete the following sentences by underlining the correct phrase in each bracket. [2]

The wave has a frequency of (7.5Hz / 7.5cm/s / 7.5cm).

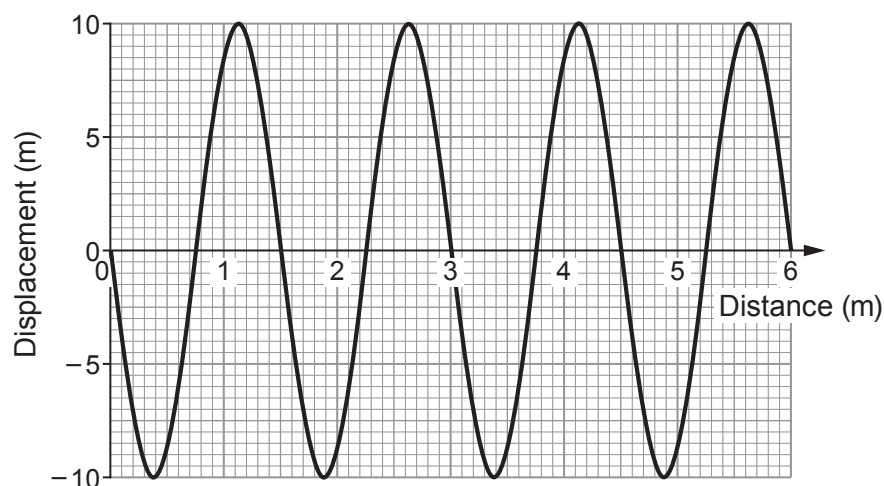
This means 7.5 waves pass a point each (second / minute / hour).

- II. Use information from parts (b)(i) and (ii) to calculate the wave speed.
Use the equation:

$$\text{wave speed} = \text{wavelength} \times \text{frequency} \quad [2]$$

Wave speed = m/s

- (c) The speed of the wave remains constant but its **frequency is doubled**.
A student draws a diagram of the expected wave. It is shown below.



Explain whether you agree with the student's diagram of the wave. [2]

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